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ENGINEERING
TEXAS A&M UNIVERSITY

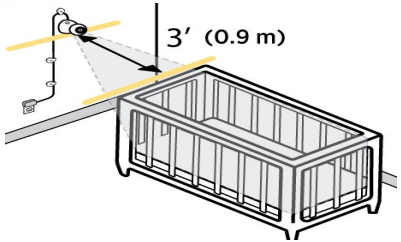
Team 64: Sound Localization Robot 4 Bi-Weekly Update 2

Samuel Bearden
Cade Christensen
Landon Harris
Xavier Valles
Sponsor: Oscar Moreria
TA: Roman Venegas

Project Summary

Problem Statement:

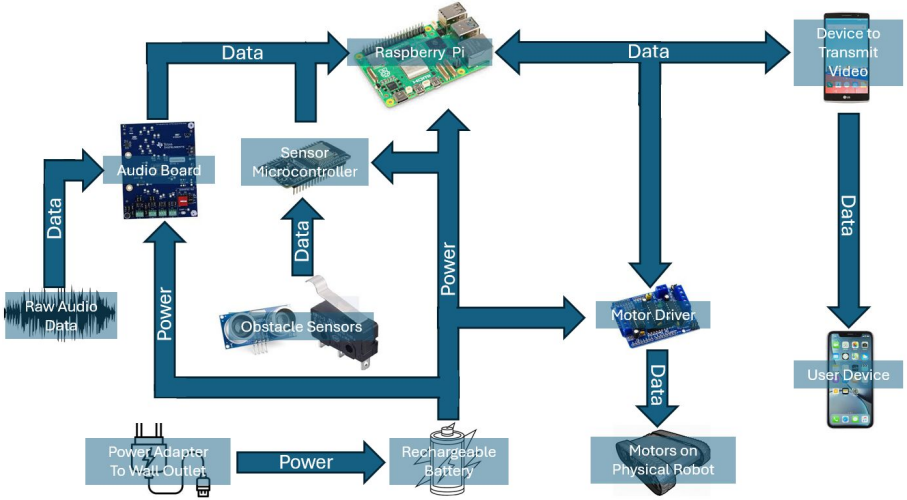
- Current baby monitors are stationary and cover only a limited area
- Some offer camera movement, but still require caretaker control



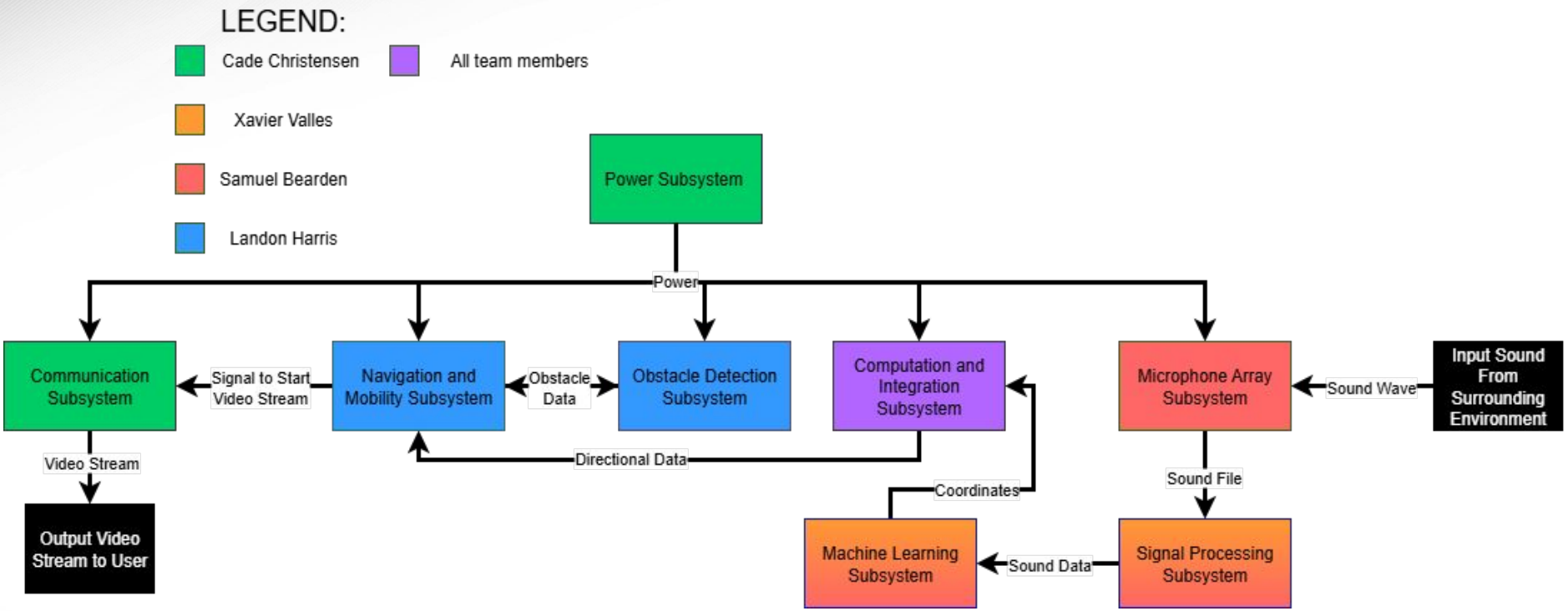
Existing monitors only cover a narrow area

Our Solution:

- A robot that autonomously navigates towards a baby's cry then using sound localization, avoiding obstacles along the way
- Starts a video call to show the baby, giving caretakers peace of mind



System Overview



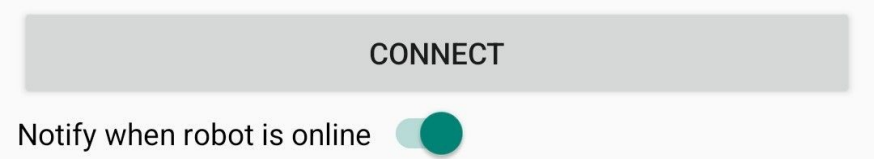
Project Timeline

Subsystem Designs and Testing (completed 1/26)	Debug Subsystem Integrations (to complete by 2/9)	Full Subsystem Integration (to complete by 2/23)	Test Autonomous Robot (to complete by 3/2)	Validations (to complete by 3/16)	Prepare Demo (to complete by 3/23)	Report (to complete by 4/6)

Power and Communication

Cade Christensen

Accomplishments since last update 12 hrs of effort	Ongoing progress/problems and plans until the next presentation
Raspberry Pi 5 HAT has been integrated Added button to turn off notifications to app Parabolic dishes integrated on robot	Continue to design modular parabolic dish Continue to work on getting the confidence score for the notification system Validate the Pi 5 HAT



Sound Characterization & Noise Filtering

Xavier Valles

Accomplishments since last update 35 hrs of effort	Ongoing progress/problems and plans until the next presentation
Dual channel voting to help with detection accuracy Uploaded all current code to Github	Ongoing problem with microphone array not recording Integrate with Sound Localization and Notification System Test model up to 65 dB using audio board and Raspberry Pi 5

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Sound Characterization & Noise Filtering

Xavier Valles

```
PS C:\Users\xitom\Documents\baby_cry_detection_2> python scripts/testing/test_multichannel_audio.py demo/rec_demo1.wav --plot
============================================================
Dual-Channel Voting Cry Detection
============================================================
Input:     demo/rec_demo1.wav
Threshold: 0.5  |  Voting: weighted  |  Plots: On
Loaded model from results/train_2025-11-24_19-10-26/evaluations/eval_2025-11-24_19-38-25/calibrated_model.pth
Duration: 9.98s  |  Channels: 4

SNR:  Ch0:   4.5 dB [P]  Ch1:   3.2 dB [S]  Ch2:   1.3 dB  Ch3:   1.7 dB

  Ch0: 5 segments (1 cry)  |  max conf: 0.7822
  Ch1: 5 segments (1 cry)  |  max conf: 0.7832
  Voting: 0.7827  |  Agreement: 99.89% (high)

============================================================
  RESULT: CRY DETECTED
  Confidence: 78.27%  |  Threshold: 50.00%
============================================================

  1. 1.50s - 4.50s
Saved: demo\rec_demo1_filtered.wav  (3.00s, 4-ch)

Generating filtering pipeline plots...

Generating visualization plots...
  Step 1: Spectral filtering...
  Step 2: Voice activity detection...
  Step 3: Spectral subtraction...
  Step 4: Computing acoustic features...
  Step 5: Getting ML predictions...

Visualization saved to: demo\filtering_visualizations\rec_demo1_ch0_filtering_analysis.png
```

Sound Localization Algorithm & Microphone Array

Samuel Bearden

Accomplishments since last update 34 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Adapted Code for model to integrate with other subsystem code • Wrote script for team to drive the whole integrated code systems • Designed 3d printed holders to mount the perf boards onto the robot • Mounted parabolic dishes, perf boards, and raspberry pi onto the robot	• Issue with TI board not responding. • Working on continued integrating of all code and testing the code together • Recapturing training data with parabolic dishes for my model

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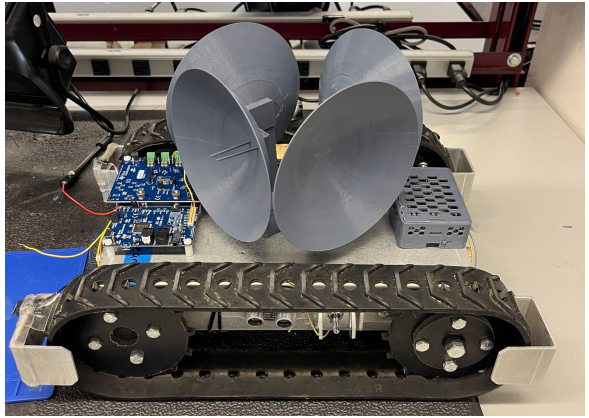
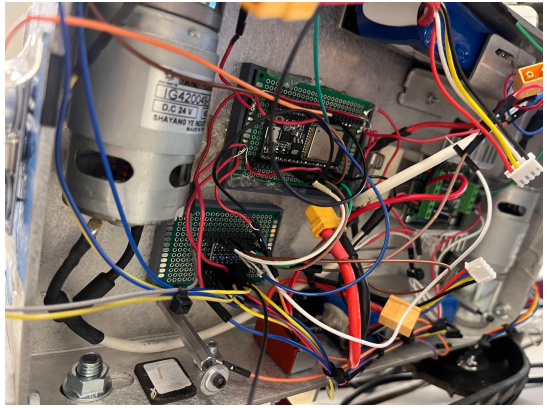
Sound Localization Algorithm & Microphone Array

Samuel Bearden

```python
def look_for_sound():
    #place holder for what will be the return of if there is a sound. if there is a sound also return the file
    pass

def process_sound_file(sound_file):
    engine = function_calls.Infer()
    result_string = engine.process_file(sound_file)
    dist = re.search(r"Distance: (\d+\.?\d*) ft", result_string)
    distance = float(dist.group(1))
    deg = re.search(r"Source \d+: (\d+\.?\d*)°", result_string)
    degrees = float(deg.group(1))
    rads=radians(degrees)
    x=distance*cos(rads)
    y=distance*sin(rads)
    return (x, y, distance)

def drive_robot(coordinates):
    # placeholder for robot algortihm that will navigate and then send back to start
    pass



def main():
    while True:
        sound_file = None
        while True:
            is_sound_detected, sound_file = look_for_sound()
            if is_sound_detected:
                break

        xyz = process_sound_file(sound_file) # returns x,y, distance in a tuple

        drive_robot(xyz)

        #    back to the start to listen again


if __name__ == "__main__":
    main()
```





Navigation/Mobility, & Obstacle Avoidance

Landon Harris

Accomplishments since last update 25 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Completed testing and debugging of obstacle avoidance algorithm, handling edge cases • Successful communication of ESP32 with Raspberry Pi in preparation for implementing sound localization algorithm	• Ongoing code to integrate obstacle avoidance communication with sound localization algorithm • Test obstacle avoidance with sound localization implementation

Navigation/Mobility, & Obstacle Avoidance

Landon Harris

Scenario	Obstacles Detected	Expected Behavior	Tested
Basic Movement	Left only	Continue straight (ignore side obstacle)	Success
Basic Movement	Right only	Continue straight (ignore side obstacle)	Success
Basic Movement	None	Continue straight	Success
Front Obstacle & Left Blocked	Front + Left	Turn right; follow obstacle on left; when left clears turn left back to original path; STOP	Success
Front Obstacle & Right Blocked	Front + Right	Turn left; follow obstacle on right; when right clears turn right back to original path; STOP	Success
Front Obstacle & Both Sides Clear	Front only	Choose side with greater clearance; follow obstacle; return to path; STOP	Success
All Sensors Blocked	Front + Left + Right	Probe right then left; turn into first clear direction; follow obstacle; STOP	Success
Obstacle Appears During FOLLOW_RIGHT	Front blocked while in FOLLOW_RIGHT state	Stop; enter REVERSE_OUT; probe for escape	Success
Obstacle Appears During FOLLOW_LEFT	Front blocked while in FOLLOW_LEFT state	Stop; enter REVERSE_OUT; probe for escape	Success
Both Probes Blocked	Front + Left + Right with both probes failing	Stop and wait (fully boxed in); STOP	Success
Narrow Corridor	Left + Right (front clear)	Continue straight without triggering avoidance	Success

Execution Plan

	1/12/2026	1/19/2026	1/26/2026	2/2/2026	2/9/2026	2/16/2026	2/23/2026	3/2/2026	3/9/2026	3/16/2026	3/23/2026	3/30/2026	4/6/2026	4/13/2026	4/20/2026	4/27/2026	Date
Status Update 1																	1/26/2026
Software Integration																	2/9/2026
Status Update 2																	2/9/2026
Sound Localization/Cry Detection Integration																	2/16/2026
Obstacle Avoidance/Sound Localization Integration																	2/23/2026
Cry Detection/Mobile App Integration																	2/23/2026
Design Blitz Participation																	2/20/2026
Status Update 3																	2/23/2026
Obstacle Avoidance/Mobile App Integration																	2/23/2026
Power System Integration with Motors and Boards																	2/23/2026
Status Update 4																	3/16/2026
Physical Design Clean Up																	3/30/2026
Status Update 5																	3/30/2026
System Validation																	3/16/2026
Prepare Demo																	3/23/2026
Trial Demo with TA																	4/9/2026
Final Design Presentation																	4/13/2026
Trial Demo with other TA																	4/16/2026
Final System Demo																	4/21/2026
Showcase																	4/24/2026
Final Report																	4/27/2026

Not Started
In Progress
Completed
Behind Schedule

Validation Plan

Paragraph #	Test Name	Success Criteria	Methodlogy	Status	Responsible Engineer(s)
3.2.1.1.	Cry Detection Accuracy	**The system correctly identifies an infant cry at least 90% of the time under typical household noise conditions. False positives remain below 10%**	Play recorded infant cry samples at varying volumes in a typical household environment with background noise. Record detection rate and false positive rate from ambient sounds.	Success	Xavier Valles
3.2.1.2.	Sound Localization Accuracy	**The system correctly localizes the sound source (room or zone) at least 90% of the time in typical indoor conditions.**	The system will be tested at different angles around the robot. A successfull test is one with a 90% accuracy in distance and direction	Success	Samuel Bearden
3.2.1.3.	Sound Localization Range	**The system correctly localizes the sound source (room or zone) from at least 10 feet away from the source**	The system will be tested at different distances up to10 feet of the microphone array. A successfull test is one with a 90% accuracy in distance and direction	Success	Samuel Bearden
3.2.1.4.	Navigation Response Time	**The system autonomously navigates to the identified room within 3 minutes of cry detection, accounting for obstacles and typical indoor conditions**	Place system in starting location. Trigger cry sound in target room. Use stopwatch to measure time from cry detection to system arrival in target room. Repeat across various room configurations	Untested	Landon Harris/Samuel Bearden
3.2.1.5	Obstacle Detection Accuracy	**The system detects obstacles in its path with at least 95% accuracy within a detection range of up to 2 meters.**	Place various obstacles at measured distances from 0.2m to 2m in system's path. Record detection events vs actual obstacles. Calculate detection accuracy percentage.	Success	Landon Harris
3.2.2.1	Mass	**The mass of the system should be ≤ 15 pounds.**	Weigh the complete assembled system including all components, batteries, and sensors using a calibrated digital scale. Record mass in both pounds and kilograms. Verify total mass does not exceed 15 lbs	Untested	Full Team
3.2.2.2	Volume Envelope	**The volume envelope of the system should be less than or equal to 293 millimeters in height, 314 millimeters in width, and 458 millimeters in length.**	Measure the complete assembled system dimensions using calipers and measuring tape. Measure maximum height, width, and length of the system including all protruding components. Verify dimensions are within specified envelope: H ≤ 293 mm, W ≤ 314 mm, L ≤ 458 mm.	Untested	Full Team
3.2.2.3	Mounting	**The system should require no permanent installation and be operable on standard household flooring.**	Place system on various standard household flooring types. Verify system operates autonomously without any mounting hardware, adhesives, or permanent fixtures. Test navigation and stability on each surface type. Confirm system can be moved and relocated by user without tools or installation procedures.	Untested	Full Team
3.2.3.1.1	Power Consumption	**The system operates without exceeding 60 W peak power under normal operation.**	Connect system to power meter/analyzer during normal operation cycle including idle, listening, navigating, and obstacle avoidance modes. Record peak power draw. Verify stays below 60W threshold.	Untested	Cade Christensen
3.2.3.1.2	Input Voltage Level	**The system operates correctly with input voltage between 22–29 VDC.**	Use variable DC power supply to test system functionality at 22V, 25.5V (nominal), and 29V. Verify all subsystems function correctly at each voltage level. Monitor for any degradation or errors.	Untested	Cade Christensen
3.2.3.2.2	Diagnostic Output	**The system correctly diagnoses errors and reports them accordingly**	The system reports errors correctly when they arise. The different errors depend of different tests, but success is determined by the ability to recognize and report alll errors	Untested	Samuel Bearden
3.2.4.1	Operating Temperature Range	**The system operates reliably within typical household temperature ranges of (68°F to 72°F)**	Place system in temperature-controlled environment. Test full functionality at 68°F, 70°F, and 72°F. Verify cry detection, localization, and navigation perform within specifications at all temperatures.	Untested	Cade Christensen
3.2.4.2	Noise Tolerance	**The cry detection system remains effective in background household noise levels up to 65 dB**	Generate controlled background noise at measured levels (50dB, 55dB, 60dB, 65dB) using typical household sounds. Play infant cry samples at each noise level. Verify detection accuracy remains ≥90%.	In Progress	Xavier Valles
N/A	Full System Demo			Untested	Full Team

Thank You For Listening!
Questions?